Uncertainty-Aware Skin Cancer Classification with Deep Ensembles

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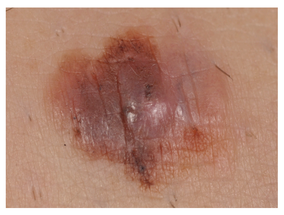
About Melanoma

Skin cancer is a major public health problem, with over 5,000,000 newly diagnosed cases in the United States every year. Melanoma is the deadliest form of skin cancer, responsible for an overwhelming majority of skin cancer deaths. In 2015, the global incidence of melanoma was estimated to be over 350,000 cases, with almost 60,000 deaths. Although the mortality is significant, when detected early, melanoma survival exceeds 95%.



About Dermoscopy

Dermoscopy is a non-invasive skin imaging technique that uses optical magnification and either liquid immersion or cross-polarized lighting to make subsurface structures more easily visible when compared to conventional clinical images.It allows for the identification of dozens of morphological features that are particularly important in identifying malignant melanoma. Melanoma is amenable to early detection by expert visual inspection or automated detection with image analysis.

Convolution Neural Network

Convolutional Neural Networks (CNNs) are a class of deep learning models designed for image analysis. CNNs have demonstrated expert-level performance in skin lesion classification tasks, making them a powerful tool for automated melanoma detection[2].

However, despite their high accuracy, CNNs often produce overconfident and poorly calibrated predictions, where predicted probabilities do not accurately reflect true likelihoods.

In this work, we use Inception v3, a deep CNN architecture pretrained on ImageNet and fine-tuned for dermoscopic image classification [1].

Uncertainty

- CNNs often produce overconfident predictions, even when incorrect.
- In high-stakes applications such as medical diagnosis, predictive accuracy alone is insufficient, as misclassification of melanoma can lead to missed detection or unnecessary treatment.
- Therefore, models must also quantify uncertainty and identify when their predictions may be unreliable.

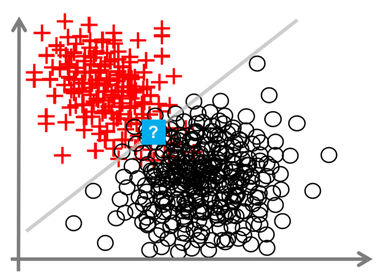
Uncertainty Quantification

Uncertainty quantification (UQ) aims to measure the reliability of a model's predictions. Uncertainty in machine learning is broadly decomposed into two components: Epistemic uncertainty and Aleatoric uncertainty.

$$\underbrace{H[y \mid x, \mathcal{D}]}_{\text{Total Predictive Uncertainty}} = \underbrace{\mathbb{E}_{\theta \sim p(\theta \mid \mathcal{D})}\big[H[y \mid x, \theta]\big]}_{\text{Aleatoric}} + \underbrace{I[y\,;\,\theta \mid x, \mathcal{D}]}_{\text{Epistemic}}$$

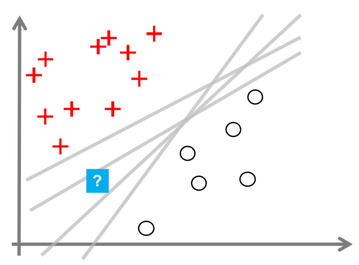
$$\underbrace{H\big[\tfrac{1}{M}\textstyle\sum_m p_{\theta_m}(y \mid \mathbf{x})\big]}_{\text{Total Predictive Uncertainty}} = \underbrace{\tfrac{1}{M}\textstyle\sum_m H[p_{\theta_m}(y \mid \mathbf{x})]}_{\text{Aleatoric}} + \underbrace{I[y\,;\,\theta \mid \mathbf{x}, \mathcal{D}]}_{\text{Epistemic}}$$

Aleatoric Uncertainty



Even with precise knowledge about the optimal hypothesis, the prediction at the query point (indicated by a question mark) is aleatorically uncertain, because the two classes are overlapping in that region [3].

Epistemic Uncertainty



A case of epistemic uncertainty due to a lack of knowledge about the right hypothesis, which is in turn caused by a lack of data [3].

Ensembles

Ensembles of NNs (Deep Ensembles [4]) combine the predictions of "**M**" independently trained models to reduce variance and improve reliability, rather than relying on a single trained model. We train Deep Ensembles consisting of M = 4 independently fine-tuned Inception v3 models initialized from ImageNet-pretrained weights.

Given models with parameters $\{\theta_m\}_{m=1}^{M}$, the ensemble predictive distribution is:

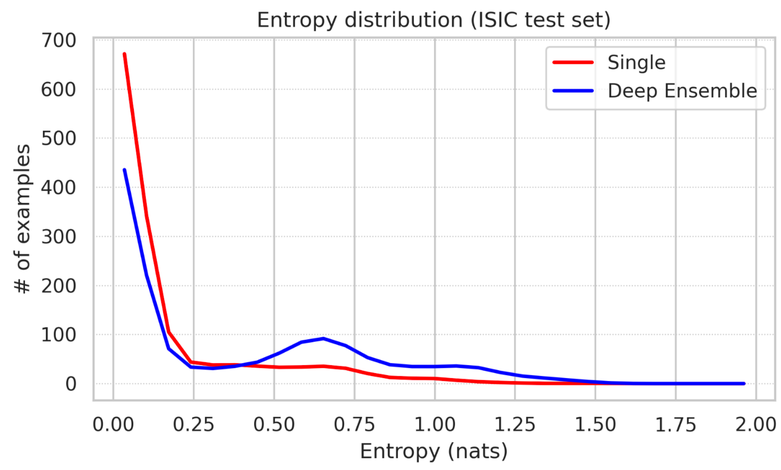
$$p_E(y \mid \mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M} p_{\theta_m}(y \mid \mathbf{x})$$

Results

We evaluate the performance of the single baseline model and the Deep Ensembles (four Inception v3 networks) on the ISIC–2018 Task 3 test set. The results are summarized in the following table:

Metric	Single (Baseline)	Deep Ensembles
Accuracy ↑	0.6635	0.6804
Negative Log-Likelihood (NLL) ↓	2.9695	1.7092
Brier Score ↓	0.6050	0.4783
Expected Calibration Error (ECE) ↓	0.2786	0.1536
Mean Predictive Entropy	0.1476	0.3763

The single baseline produces very low entropy values for most samples, indicating that it tends to make highly confident predictions. In contrast, Deep Ensembles show a wider spread of entropy values, including higher-entropy regions that correspond to greater predictive uncertainty. The predictive entropy distributions for the two models are shown in the figure below.



Conclusion

Deep Ensembles consistently outperform the single-model baseline in the skin lesion classification task, offering improved accuracy and substantially better calibration. Importantly, they more reliably identify ambiguous or uncertain cases, highlighting their advantage in settings where incorrect high-confidence predictions may lead to harmful clinical outcomes.

References

[1] Codella et al., Skin Lesion Analysis Toward Melanoma Detection (ISIC 2018), arXiv, 2019.

[2] Esteva et al., Dermatologist-Level Classification of Skin Cancer with Deep Neural Networks, Nature, 2017.

[3] Hüllermeier & Waegeman, Aleatoric and Epistemic Uncertainty in Machine Learning, Machine Learning, 2021.

[4] Lakshminarayanan et al., Deep Ensembles for Uncertainty Estimation, NeurIPS, 2017.